

Machine Learning-based Bengali Text Sentiment Analysis Framework

Report on Thesis Proposal



Course No: CSE 061 4100

Course Title: Project and Thesis - I

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Date of Submission: September 18, 2026

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1. Introduction

The rapid expansion of localized digital platforms has created a critical need for automated text analytics in regional languages. Standard global models optimized for English fail on Bengali script due to its morphological richness, complex inflections, phonetic variations, and low-resource status (Hoque, Sultana, & Rasel, 2026). Sentiment and sarcasm detection in Bangla remain uniquely challenging because of these distinct linguistic properties (Hoque, Sultana, & Rasel, 2026). Categorizing public opinion into clean, actionable dimensions requires architectures engineered specifically around the structural realities of Bengali web speech rather than adapted English frameworks.

1.1 Motivation

Automated monitoring across Bengali social media, customer service, and public feedback channels is essential to process high volumes of unstructured colloquial text. Inaccurate sentiment interpretation leads to direct financial loss, degraded user trust, and unaddressed public feedback. Existing models struggle with widespread Bangla-English code-mixing, transliteration variations, and intra-sentential language switching (Alam, Chowdhury, Ahmed, Abrar, & Haque, 2026). Furthermore, native sarcasm frequently triggers misinterpretation of humor-based comments (Biswas, et al., 2025). Resolving these regional text patterns is crucial for establishing robust, production-ready multilingual NLP tools that genuinely serve linguistically diverse communities.

1.2 Problem Statement

Current Bengali sentiment engines struggle with contextual ambiguity, code-switching, and severe class imbalance. Benchmark evaluations reveal that standard models systematically misclassify neutral and negative samples as positive due to majority-class vocabulary dominance (Kabir, Mahfuz, Raiyan, Mahmud, & Hasan, 2023, p. 1240). This issue is compounded by the structural heterogeneity of neutral expressions (Mudinas, Zhang, & Levene, 2018, p. 280) and the fact that over 42% of negative instances in code-mixed Bangla-English text exhibit sarcastic characteristics (Alam, Chowdhury, Ahmed, Abrar, & Haque, 2026). This study addresses these bottlenecks by establishing a standardized, mathematically balanced mapping across target labels: Negative (0), Positive (1), and Neutral (2).

1.3 Objectives

- **Preprocessing & Annotation Pipeline:** Develop a reproducible Bengali text-preprocessing pipeline—encompassing Unicode normalization, spelling management, and subword tokenization—paired with a standardized multi-class label mapping (Negative: 0, Positive: 1, Neutral: 2).
- **Contextual Representation & Linguistic Modeling:** Evaluate dense contextual embeddings against surface-level lexical features to determine how effectively they resolve complex Bengali language properties, including code-mixing, inflection, negation, and sarcasm.
- **Class Imbalance Optimization:** Mitigate majority-class bias and neutral/negative misclassifications through focal loss, class-weight calibration, and ablation experiments.
- **Evaluation & Robustness Assessment:** Benchmark model performance, error patterns, calibration, and out-of-domain transfer across diverse Bengali social media and review datasets using standard classification metrics and confusion matrices.

2. Related Works

Research on sentiment and emotion classification has progressed from psychologically grounded emotion models and manually constructed lexical resources to large-scale supervised learning and contextual transformer architectures. Early theoretical work conceptualized emotions as a limited set of basic categories, establishing an important foundation for distinguishing affective states in language (Ekman, 1992). Building on this perspective, crowdsourced emotion lexicons demonstrated that words and short expressions can be associated with multiple emotional dimensions through human annotation, providing valuable resources for feature-based sentiment and emotion analysis (Mohammad & Turney, 2012). However, lexicon-based approaches are limited when sentiment depends on negation, discourse context, sarcasm, code-mixing, or culturally specific expressions.

The development of distant-supervision methods enabled sentiment classifiers to be trained from large collections of weakly labeled social-media data. Go et al. used emoticons as noisy sentiment indicators for automatically constructing training data, demonstrating the potential of large-scale supervision for Twitter sentiment classification (Go, Bhayani, & Huang, 2009). Although this approach substantially reduces annotation costs, automatically generated labels may contain considerable noise and may not adequately represent neutral, ambiguous, or context-dependent language. These limitations are especially important for Bengali social-media text, where spelling variation, transliteration, informal constructions, and mixed-language expressions can obscure the intended polarity.

Transformer-based language models have subsequently improved the representation of contextual meaning. BERT introduced bidirectional pre-training through masked language modeling and next-sentence prediction, enabling downstream classifiers to model relationships between words and their surrounding contexts (Devlin, Chang, Lee, & Toutanova, 2019). Such contextual representations are relevant to Bengali sentiment analysis because a single lexical item may express different sentiments depending on negation, intensification, syntactic structure, or the broader discourse. Nevertheless, the performance of a general-purpose transformer depends strongly on the quality and representativeness of its training data, tokenizer, domain adaptation, and task-specific annotation scheme.

More recent work has moved beyond coarse positive–negative classification toward fine-grained and multi-label emotion analysis. GoEmotions, for example, provides a large collection of text annotated with a broad set of emotion categories, illustrating the value of detailed affective taxonomies for capturing distinctions that are lost in binary or ternary sentiment schemes (Demszky, et al., 2020). This direction is relevant to the present study because Bengali statements that appear neutral at the polarity level may express uncertainty, disappointment, amusement, or restrained criticism. A richer understanding of such distinctions can support more reliable annotation guidelines and error analysis, even when the final task uses the three target classes Negative, Positive, and Neutral.

The neutral class remains a persistent challenge in sentiment classification. Evidence from Twitter sentiment experiments shows that neutral documents are not homogeneous: some contain balanced positive and negative content, whereas others express no discernible sentiment (Mudinas, Zhang, & Levene, 2018, p. 280). Consequently, a classifier trained with a single neutral label may struggle to learn a coherent decision boundary. This problem is also evident in Bengali sentiment datasets, where models frequently perform poorly on the Neutral

class and tend to misclassify negative and neutral instances as Positive (Kabir, Mahfuz, Raiyan, Mahmud, & Hasan, 2023, p. 1240). The dominance of positive vocabulary in imbalanced data further increases this bias, indicating that overall accuracy alone may conceal substantial weaknesses in minority-class recognition (Kabir, Mahfuz, Raiyan, Mahmud, & Hasan, 2023, p. 1240).

Sarcasm adds another layer of difficulty because the literal polarity of an utterance may conflict with its intended meaning. BanglaSarc3 frames sarcasm detection as a distinct Bengali NLP problem and provides a balanced ternary resource containing sarcastic, neutral, and non-sarcastic Facebook comments (Biswas, et al., 2025). This work emphasizes the importance of context-aware modeling and careful distinction between neutrality and non-sarcastic language. In addition, recent research on Bangla–English code-mixed text reports that more than 42% of negative sentiment instances in an external dataset exhibit sarcastic characteristics, suggesting that sarcasm is closely intertwined with polarity in multilingual online communication (Alam, Chowdhury, Ahmed, Abrar, & Haque, 2026). Transliteration variation, language switching, cultural references, and implicit meaning therefore need to be considered when developing Bengali sentiment classifiers.

Existing studies also demonstrate that domain-specific data and model adaptation are essential for robust performance. A bootstrap approach to domain-specific sentiment classification reports that pseudo-labeled data can support classification, but also shows that the heterogeneous nature of neutral language makes representative training examples difficult to obtain (Mudinas, Zhang, & Levene, 2018, p. 280). Similarly, the BanglaBook findings indicate that class imbalance and overlapping lexical distributions can produce systematic misclassification, particularly when positive vocabulary dominates the corpus (Kabir, Mahfuz, Raiyan, Mahmud, & Hasan, 2023, p. 1240). These observations motivate the use of stratified data partitioning, class-weight calibration, focal loss, class-specific evaluation, and targeted error analysis in the proposed framework.

Despite these advances, an integrated Bengali framework that jointly addresses script variation, contextual polarity, neutral-class ambiguity, sarcasm, code-mixing, and class imbalance remains necessary. The present study responds to this gap by adopting a transparent three-class mapping—Negative, Positive, and Neutral—while evaluating not only aggregate performance but also macro-averaged metrics, confusion matrices, calibration, and class-specific errors. In contrast to approaches that rely solely on surface lexical features or overall accuracy, the proposed methodology combines script-aware preprocessing, subword tokenization, contextual embeddings, and imbalance-aware optimization. This design is intended to produce a more reproducible and practically useful Bengali sentiment-analysis system for review, social-media, customer-feedback, and public-service applications. Furthermore, this research integrates concept-level sentiment analysis to explicitly model negation and affective modifiers, which enhances interpretability compared to traditional black-box deep learning models.

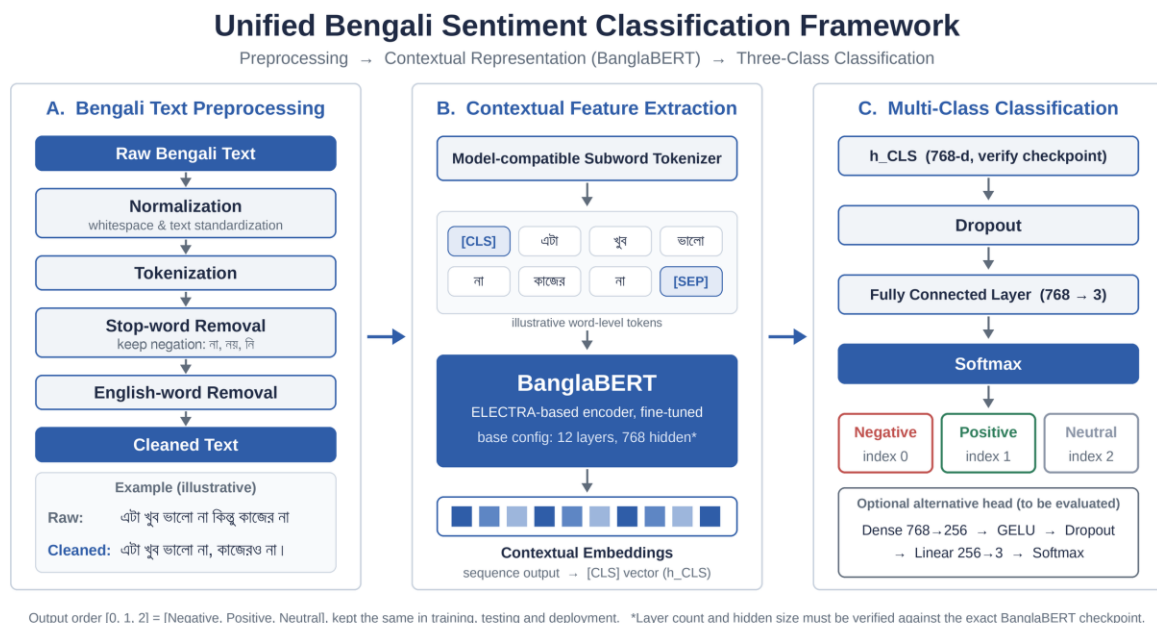
The literature matrix below details the computational frameworks, dataset configurations, and evaluation metrics extracted from foundational studies within text categorization and localized sentiment analysis:

Table 2.1: Comparative Summary of Related Studies in Sentiment and Emotion Analysis

Author Name	Paper Title	Methods	Accuracy / Target Mapping	Reference
Demszky et al.	GoEmotions: A dataset of fine-grained emotions	BERT-base Fine-tuning with Multi-label Sigmoid	46% Average Macro F1-score	(Demszky, et al., 2020)
Devlin et al.	BERT: Pre-training of deep bidirectional transformers for language understanding	Masked Language Modeling (MLM) & NSP	80.5% GLUE Benchmark	(Devlin, Chang, Lee, & Toutanova, 2019)
Ekman, P.	An argument for basic emotions	Theoretical Framework of Discrete Archetypes	Qualitative Psychological Baseline	(Ekman, 1992)
Go et al.	Twitter sentiment classification using distant supervision	Emoticon Distant Supervision (Naive Bayes, SVM)	> 80% Accuracy on Polarized Text	(Go, Bhayani, & Huang, 2009)
Mohammad & Turney	Crowdsourcing a word-emotion association lexicon	Amazon Mechanical Turk Crowdsourcing Layout	>90% Inter-annotator Agreement	(Mohammad & Turney, 2012)

3. Proposed Methodology

The proposed research develops a Bengali text sentiment classification framework using Bengali text preprocessing, contextual feature extraction, and multi-class sentiment prediction. The framework classifies Bengali text into three categories: Positive, Negative, and Neutral. The methodology consists of dataset collection and annotation, preprocessing, dataset partitioning, contextual representation using BanglaBERT, multi-class classification, and model evaluation.



Overall Methodological Workflow

1. Raw Bengali Text Dataset (18,000 annotated samples)
2. Data preprocessing: tokenization → stop-word removal → English word removal
3. Stratified dataset split: 80% training / 10% validation / 10% testing
4. Comparative models: machine learning, deep learning, and transformer baselines
5. BanglaBERT-compatible subword tokenization and contextual embedding
6. Three-class Softmax classification: Negative (0), Positive (1), Neutral (2)
7. Evaluation: accuracy, precision, recall, macro-F1, and confusion matrix

3.1 Dataset Description and Annotation

The proposed framework will use a Bengali text dataset containing 18,000 annotated samples. Text may be collected from relevant Bengali-language sources, such as public social media discussions, online reviews, and other textual resources, subject to data availability and applicable permissions. Each record will contain a serial number, Bengali text, and a sentiment annotation.

3.1.1 Dataset Distribution

Table 3.1: Distribution of Sentiment Classes in the Proposed Dataset

Sentiment class	Label	Samples	Percentage
Positive	1	7,000	38.89%
Negative	0	7,000	38.89%
Neutral	2	4,000	22.22%
Total	—	18,000	100%

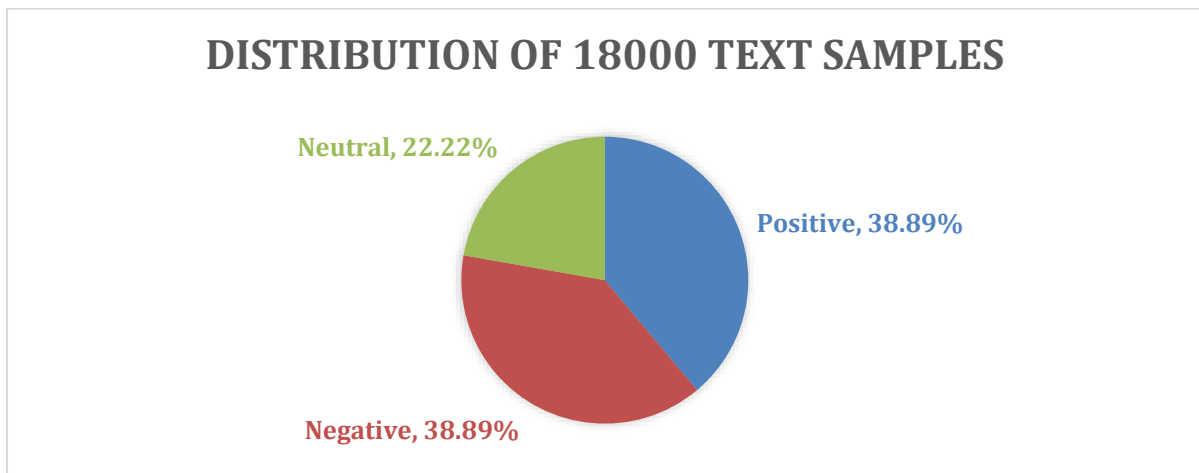


Figure 3.1. Distribution of the three sentiment classes in the proposed dataset.

Positive and Negative each account for 38.89% of the dataset, while Neutral accounts for 22.22%. The dataset is therefore moderately imbalanced, and evaluation should report class-specific performance as well as overall accuracy.

3.1.2 Dataset Information and Statistical Metrics

Table 3.2: Dataset Information and Statistical Metrics

Metric	Description / reporting method
Total samples	18,000 annotated records
Total tokens	Sum of token counts across all preprocessed text samples; calculate from the dataset
Unique tokens	Number of distinct tokens under the stated tokenizer; calculate from the dataset
Mean tokens per sentence	Average token count per sample: $\mu L = (\sum L_i) / N$
Median tokens per sentence	Median of the per-sample token-count distribution
Maximum tokens per sentence	Largest per-sample token count

Token statistics must be calculated from the actual dataset after applying the documented preprocessing and tokenization rules. They cannot be inferred from class counts alone.

3.1.3 Data Labeling Format

Table 3.3: Data Labeling Format for Sentiment Classification

Serial Number	Bangla Text	Sentiment Analysis	Label
1	খুলনার ৫টি কলেজে বইমেলায় আয়োজন নিঃসন্দেহে একটি ভালো উদ্যোগ।	Positive	1
2	খুলনা থেকে ঢাকাগামী জাহানাবাদ এক্সপ্রেস ট্রেন এর পরিবেশ বেশ অপরিষ্কার এবং সেবার মান খারাপ।	Negative	0
3	আজ গণভবনে শিক্ষামন্ত্রী পাঠ্যপুস্তকের মান পরিবর্তন নিয়ে উর্ধ্বতন কর্মকর্তাদের সাথে বৈঠক করেছেন।	Neutral	2
4	প্রাণ কোম্পানির নতুন বাজারজাত করা দেশি ঘি এর গুণগত মান বেশ ভালো।	Positive	1
5	দীর্ঘদিন ধরে আবেদন করা সত্ত্বেও এখানকার ভাঙা রাস্তা মেরামত করা হয়নি, যা অত্যন্ত নিন্দনীয়।	Negative	0

The examples are illustrative; final labels must reflect each text's meaning and context.

Label 1 represents Positive sentiment (satisfaction, appreciation, approval, or favorable opinion). Label 0 represents Negative sentiment (dissatisfaction, criticism, anger, or unfavorable opinion). Label 2 represents Neutral sentiment (factual statements without a clear positive or negative orientation). Each sample will receive one label, with ambiguous cases reviewed using a consistent annotation guideline.

3.2 Bengali Text Preprocessing

Preprocessing is the first computational stage. It reduces irrelevant text noise and prepares Bengali text for classification. The pipeline explicitly includes tokenization, stop-word removal, and English word removal.

3.2.1 Tokenization

Each Bengali text sample will be divided into individual word-level tokens using a Bengali-compatible tokenizer. For example, “অসাধারণ একটি উদ্যোগ।” may be tokenized as [“অসাধারণ”, “একটি”, “উদ্যোগ”]. Punctuation handling will be applied consistently.

3.2.2 Stop-Word Removal

A selected list of frequent Bengali functional words (for example, এবং, অথবা, কিন্তু, যে, এই, একটি) may be removed when they contribute little to the task. The list will be documented and applied consistently. Polarity-bearing negation terms such as না, নয়, and নি must be retained because removing them can reverse sentiment.

3.2.3 English Word Removal

English-language tokens will be removed using Unicode-aware rules that identify English alphabetic words. For example, “সার্ভিসটি খুব good কিন্তু delivery late” may become “সার্ভিসটি খুব কিন্তু”. Because code-mixed English words can carry sentiment, the effect of this step should be examined during evaluation and reported as a methodological limitation.

3.2.4 Preprocessing Sequence

1. Load the raw Bengali text and its annotation.
2. Normalize unnecessary whitespace and standardize text representation.
3. Tokenize the text into word-level units.
4. Remove selected Bengali stop words while preserving negation.
5. Remove English-language words.
6. Retain the cleaned text or token sequence for model input.
7. Preserve the original serial number, text, and label for traceability.

3.3 Dataset Partitioning and Training Architecture

The preprocessed dataset will be divided using stratified sampling. The requested 80% training allocation will be retained, while the remaining 20% will be split equally into validation and testing subsets. This separation supports model tuning without using the final test set.

Table 3.4: Dataset Partitioning for Training, Validation, and Testing

Partition	Percentage	Samples
Training	80%	14,400
Validation	10%	1,800
Testing	10%	1,800
Total	100%	18,000

3.3.1 Split Distribution by Sentiment

Table 3.5: Sentiment-wise Distribution Across Dataset Subsets

Sentiment	Training	Validation	Testing
Positive	5,600	700	700
Negative	5,600	700	700
Neutral	3,200	400	400
Total	14,400	1,800	1,800

Stratification preserves approximately the original class proportions in each subset. Duplicate or near-duplicate texts should be identified before splitting; where appropriate, related comments from the same source or discussion may be grouped to reduce data leakage.

3.4 Models for Comparative Evaluation

To examine whether contextual representations improve Bengali sentiment classification, three groups of models will be compared: classical machine learning, deep learning, and transformer-based models. All models will use the same dataset, the same stratified 80/10/10 split, and the same evaluation metrics. Table 3.6 summarizes the proposed models.

Table 3.6: Proposed Models for Comparative Evaluation

Group	Proposed models	Input representation	Role
Machine Learning	Logistic Regression, Linear SVM, Multinomial Naive Bayes	TF-IDF (word unigrams and bigrams)	Lexical baseline
Deep Learning	BiLSTM, CNN-BiLSTM	Trainable word embeddings	Sequence models without pretrained context
Transformer	BanglaBERT (primary); mBERT and XLM-R (optional)	Model-specific subword tokenizer	Contextual representation

3.4.1 Machine Learning Models

Logistic Regression, Linear Support Vector Machine (SVM), and Multinomial Naive Bayes will be trained on TF-IDF features built from word unigrams and bigrams. These models treat words as independent surface features. They are fast, easy to reproduce, and widely used in earlier

Bengali sentiment studies (Haque, Islam, Tasneem, & Das, 2023). They form the lexical baseline for Objective 2, which compares contextual embeddings against surface-level lexical features.

3.4.2 Deep Learning Models

BiLSTM and CNN-BiLSTM models will be trained on trainable word embeddings. They learn word order and local patterns from the training data without large-scale pretraining. Similar architectures have been used for under-resourced Bengali text (Karim, Chakravarthi, McCrae, & Cochez, 2020). These models show how much sequence modeling alone contributes before contextual pretraining is introduced.

3.4.3 Transformer Models

BanglaBERT is the primary model and will be fine-tuned as described in Sections 3.5 to 3.7. Multilingual transformers such as mBERT and XLM-R may be added as optional comparison models to check whether a Bengali-specific model performs better than multilingual ones. Fine-tuned transformers have been applied to Bengali multi-class sentiment analysis in earlier work (Tonmoy, 2023).

3.4.4 Fair Comparison Protocol

All models will be trained on the training subset and tuned on the validation subset. Class weighting will be applied to every model that supports it. Models will be compared mainly by Macro-F1 and class-wise results on the same test subset. No model is assumed to be superior before the experiments are performed. Because stop-word and English word removal may affect transformer models differently from ML and DL models, the effect of preprocessing should be examined during evaluation.

3.5 Contextual Feature Extraction Using BanglaBERT

The preprocessed Bengali text will be converted into contextual numerical representations using a pretrained BanglaBERT checkpoint. BanglaBERT is based on the ELECTRA architecture and will be fine-tuned for the three-class sentiment task.

3.5.1 Model Input Representation

The cleaned text will be passed to the tokenizer associated with the selected BanglaBERT checkpoint. The tokenizer converts text into model-compatible subword tokens and numerical IDs, and prepares special tokens, attention masks, and padding as required.

For an input sequence X , the contextual hidden representations are $H = \text{BanglaBERT}(X)$. A sequence-level representation, typically the final hidden state corresponding to the classification token, will be used by the classification head. The standard base configuration commonly uses a 768-dimensional hidden representation; this must be verified against the exact checkpoint.

The requested word-level preprocessing tokenizer and BanglaBERT's internal subword tokenizer serve different purposes. The first supports preprocessing operations; the model tokenizer ensures compatibility with the pretrained vocabulary. The same model tokenizer and label mapping must be used during training and inference.

3.6 Multi-Class Sentiment Classification

The sequence-level contextual representation will be passed to a classification head consisting of dropout regularization and a fully connected output layer. A denser head with a GELU activation may be evaluated as an alternative.

Let h_{CLS} be the contextual representation. The logits are $z = Wh_{CLS} + b$, where W and b are trainable parameters. The three logits are converted to probabilities using Softmax: $P(y=c|X) = \exp(z_c) / \sum_j \exp(z_j)$. The predicted class is the one with the highest probability.

3.6.1 Numerical Label Mapping

Table 3.7: Numerical Label Mapping for Sentiment Classes

Dataset label	Sentiment
0	Negative
1	Positive
2	Neutral

The output-index order will be explicitly defined as $[0, 1, 2] = [\text{Negative}, \text{Positive}, \text{Neutral}]$. This mapping will remain consistent across training, validation, testing, and deployment.

3.7 Model Training and Optimization

The BanglaBERT backbone and classification head will be fine-tuned using the training subset. AdamW will be used as the optimizer. Learning rate, batch size, dropout, and epoch count will be selected using validation performance.

Table 3.8: Proposed Hyperparameter Configuration for Model Training

Hyperparameter	Proposed initial setting
Pretrained model	BanglaBERT (exact checkpoint to be specified)
Training samples	14,400
Validation samples	1,800
Test samples	1,800
Optimizer	AdamW
Initial learning rate	2×10^{-5} (starting value)
Batch size	16
Maximum epochs	5
Loss function	Cross-entropy; compare with class-weighted cross-entropy
Model selection	Validation macro-F1 and/or validation loss
Early stopping	Based on validation performance

These are proposed starting values, not experimental findings. They may be adjusted according to computing resources and validation results.

The settings in Table 3.8 apply to BanglaBERT. The machine learning and deep learning baselines will be trained on the same training subset, tuned on the same validation subset, and evaluated on the same test subset, using their own hyperparameter settings.

3.8 Model Evaluation

The final trained model will be evaluated using an independent test subset consisting of 1,800 samples that were not used during the training process. This evaluation will determine how effectively the model generalizes to unseen data and correctly classifies the three sentiment categories: Positive, Negative, and Neutral. Both overall performance and class-specific performance will be reported using standard classification metrics, including Accuracy, Precision, Recall, and F1-Score. A confusion matrix will also be generated to provide a detailed view of the model's correct and incorrect predictions.

3.8.1 Accuracy

Accuracy measures the overall proportion of predictions that the model classifies correctly out of all test samples. It provides a general indication of the model's overall predictive performance.

$$\text{Accuracy} = \frac{\text{Number of Correct Predictions}}{\text{Total Number of Predictions}}$$

A higher accuracy indicates that the model correctly classifies a larger proportion of the test samples. However, accuracy alone may not fully represent performance when the classes are imbalanced.

3.8.2 Precision

Precision measures how many of the samples predicted as a particular class actually belong to that class. It is particularly useful for determining how reliable the model's positive predictions are for each sentiment category.

$$\text{Precision} = \frac{TP}{TP + FP}$$

Here, TP (True Positive) represents samples correctly predicted as the target class, while FP (False Positive) represents samples incorrectly predicted as that class. A high precision indicates fewer incorrect predictions for a particular class.

3.8.3 Recall

Recall, also known as sensitivity, measures the proportion of actual samples belonging to a particular class that are correctly identified by the model. It indicates how effectively the model detects instances of each sentiment category.

$$\text{Recall} = \frac{TP}{TP + FN}$$

Here, FN (False Negative) represents samples that actually belong to the target class but are incorrectly classified as another class. A high recall indicates that the model successfully identifies most of the actual instances of that class.

3.8.4 F1-Score

F1-Score is the harmonic mean of Precision and Recall. It provides a balanced measure when both correctly identifying a class and avoiding incorrect predictions are important.

$$F1 = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

The F1-Score ranges from 0 to 1, where a higher value indicates better performance. In this study, Macro-F1 will be emphasized because it calculates the F1-Score independently for Positive, Negative, and Neutral classes and then gives each class equal importance, regardless of the number of samples in each class.

3.8.5 Confusion Matrix

A confusion matrix will be generated to provide a detailed representation of the model's classification results across the three sentiment categories. It shows the number of samples that were correctly classified as Positive, Negative, or Neutral, as well as the samples that were misclassified into another category. This helps identify which sentiment classes the model handles effectively and which classes are more frequently confused with one another.

3.9 Summary of the Proposed Methodological Contributions

1. Structured Bengali dataset: 18,000 annotated samples with explicit Positive, Negative, and Neutral labels.
2. Bengali text preprocessing: tokenization, stop-word removal, and English word removal, with sentiment-bearing negation retained.
3. Stratified training architecture: 80% training, 10% validation, and 10% testing.
4. Comparative evaluation of machine learning, deep learning, and transformer models on the same data split.
5. Contextual feature extraction using BanglaBERT and its compatible subword tokenizer.
6. Three-class Softmax prediction with a consistent numeric label mapping.
7. Comprehensive evaluation using accuracy, class-wise precision and recall, macro-F1, and a confusion matrix.

Together, these components establish a reproducible experimental framework for Bengali sentiment classification. The effectiveness of preprocessing choices, contextual language modeling, and class-weighting strategies will be assessed empirically using the held-out validation and test subsets.

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